

Customer Retention Analysis

Predicting Repeat Purchases in Brazilian E-commerce

Business Case Study

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Project positioning

This report is written as a business-oriented case study for a portfolio. It summarizes how predictive analytics can support customer retention decisions in an e-commerce marketplace.

1. Executive Summary

E-commerce companies invest significant resources in acquiring new customers, yet long-term profitability depends largely on retaining existing ones. In marketplace businesses, a low repeat-purchase rate can reduce customer lifetime value, increase dependence on paid acquisition, and limit scalable growth.

This project analyzes customer behavior using the Olist Brazilian E-commerce dataset, which contains transactional information from orders placed between 2016 and 2018. The business objective was to identify customers with a high risk of not making a repeat purchase and translate that prediction into practical retention strategies.

The analysis combined data cleaning, feature engineering, exploratory analysis, machine learning classification, and sentiment analysis based on customer reviews. Several models were evaluated, including Decision Trees, Random Forest, K-Nearest Neighbors, and Support Vector Machines. The modeling process was designed not only to maximize technical performance, but also to support a realistic business decision: deciding which customers should receive retention actions.

The results suggest that shipping cost, product price, and delivery-related variables are relevant factors associated with customer behavior. In addition, review sentiment provides qualitative insight into pain points that are not fully captured by transactional variables alone.

The main business recommendation is to use predictive analytics as a targeting layer for retention campaigns. Instead of offering generalized discounts, Olist could prioritize high-risk customers with shipping incentives, personalized coupons, loyalty benefits, post-purchase communication, and product recommendations. This would improve marketing efficiency while supporting long-term customer lifetime value.

Business Insight

Retention is not only a marketing problem. In this project, logistics-related variables appear central to customer behavior, which means that shipping policy can directly influence loyalty and long-term profitability.

2. Business Problem

Customer acquisition is expensive. For an e-commerce marketplace, attracting a first-time buyer requires advertising, platform investment, seller management, logistics coordination, and customer service. If a large share of customers make only one purchase, the business becomes overly dependent on constantly acquiring new users.

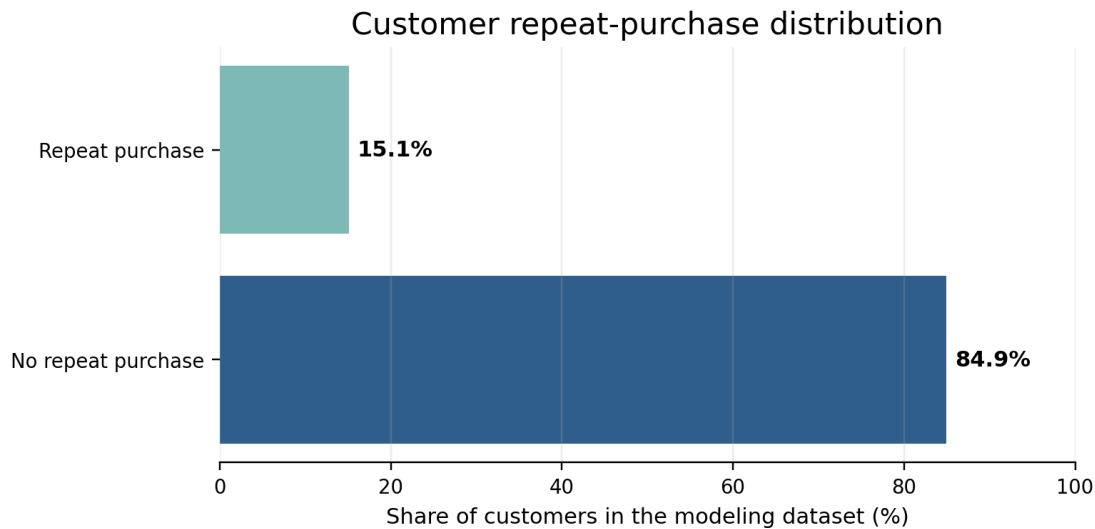


Figure 1. Repeat-purchase distribution in the modeling dataset.

The key business challenge is therefore not simply to predict whether a customer will return, but to identify which customers are at risk of not returning early enough to act. A useful model should help the company allocate retention resources more efficiently and avoid spending equally on customers with very different probabilities of repeat purchase.

From a business perspective, the most costly error is missing a customer who is unlikely to return and therefore receives no retention action. For that reason, the customer retention model should prioritize reducing false negatives for the at-risk segment, while still monitoring precision to avoid inefficient marketing spending.

Dimension	Project Decision
Business objective	Increase repeat purchases and customer lifetime value.
Analytical objective	Predict customers at risk of not making a repeat purchase.
Primary business risk	Failing to identify customers who need retention actions.
Decision supported	Who should receive shipping incentives, coupons, loyalty benefits, or post-purchase communication?

Managerial Implication

The model should not be evaluated only as a statistical exercise. Its value depends on whether it helps managers prioritize customers and design more efficient retention campaigns.

3. Data Overview

The project uses the public Olist Brazilian E-commerce dataset, which includes order, customer, payment, product, seller, delivery, and review information. The original data was distributed across multiple CSV files, requiring data integration before any modeling could be performed.

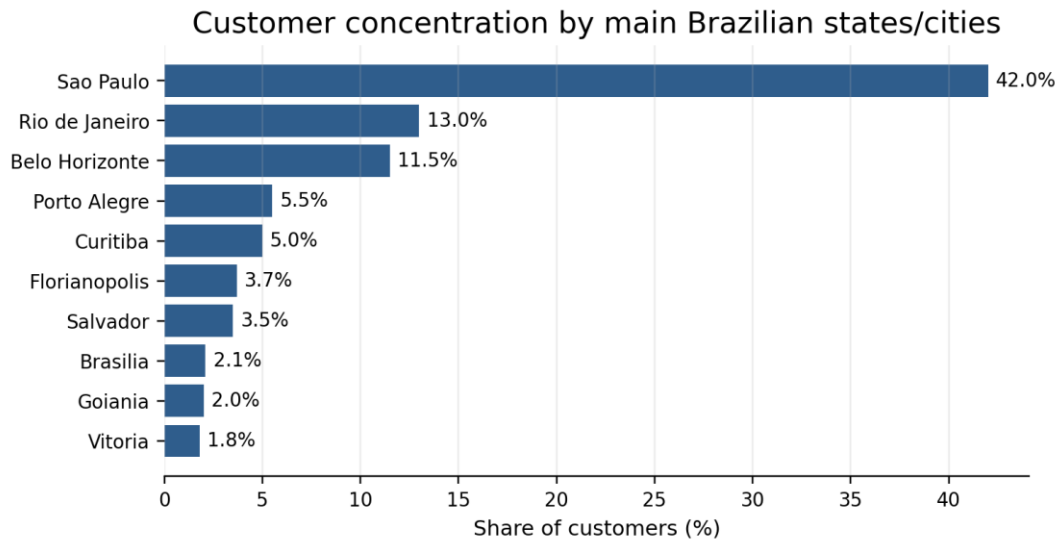


Figure 2. Customer concentration by main Brazilian state/city.

The dataset covers orders from 2016 to 2018 and provides a strong base for analyzing the customer journey from purchase to delivery and post-purchase review. The analysis focused on transforming fragmented operational data into a unified customer-level analytical dataset.

Several variables were engineered or transformed to improve analytical value. Product dimensions were summarized into product volume, payment and logistics variables were consolidated, and categorical information was adapted for machine learning. Missing values, duplicated records, outliers, and inconsistent data types were also addressed.

Dimension	Project Decision
Data source	Olist Brazilian E-commerce public dataset.
Time period	Orders placed between 2016 and 2018.
Main entities	Customers, orders, products, payments, sellers, deliveries, and reviews.
Business unit of analysis	Customer purchase behavior and likelihood of repeat purchase.
Main analytical challenge	Integrating multiple CSV files into one consistent modeling dataset.

4. Methodology

The methodology followed a business analytics workflow: define the business question, prepare the data, explore patterns, create predictive features, train classification models, evaluate results, and convert findings into recommendations.

The original datasets required cleaning and integration before modeling. Missing numerical values were imputed using robust criteria, categorical variables were transformed, duplicated records were removed, and outliers were treated to avoid distortions in model training.

Because repeat purchases were not evenly distributed across customers, class imbalance was a central modeling issue. The project therefore applied balancing techniques and stratified sampling to preserve class proportions across training and test sets.

Model evaluation was aligned with business purpose. For customer retention, the goal was to reduce the probability of missing at-risk customers. For review sentiment analysis, the priority was to correctly identify negative reviews while minimizing false positive classifications for the negative class.

Dimension	Project Decision
Step 1	Data integration from multiple CSV files.
Step 2	Data cleaning, missing value treatment, outlier analysis, and type correction.
Step 3	Feature engineering for customer, product, payment, and logistics variables.
Step 4	Class balancing and stratified train-test split.
Step 5	Model training and evaluation.
Step 6	Translation of model outputs into business recommendations.

Business Insight

A technically accurate model is not enough. The modeling strategy must reflect the economic cost of each classification error and the type of decision the company will make with the prediction.

5. Key Findings

Shipping cost is a critical retention driver

Shipping cost emerged as one of the most relevant variables in the classification process. This suggests that delivery costs may influence whether a first-time customer decides to buy again.

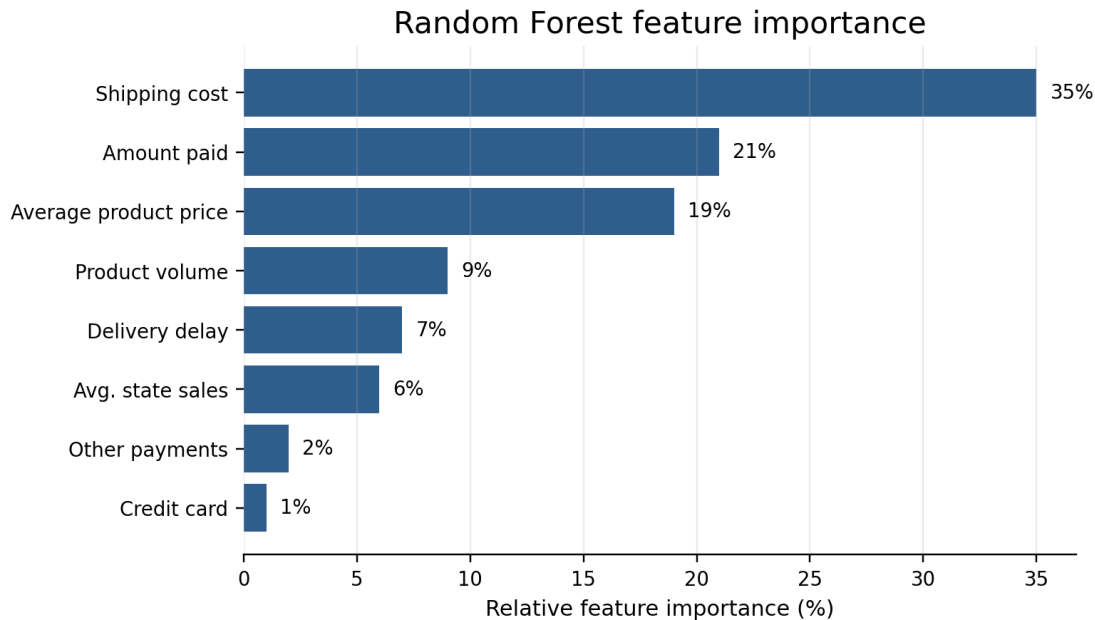


Figure 4. Random Forest feature importance, reformatted for business interpretation.

Price and logistics interact with customer behavior

Product price and shipping cost should not be analyzed separately. Customers may evaluate the total cost of the transaction, including the product price, freight, delivery time, and perceived convenience.

Delivery experience affects loyalty

Delays, delivery performance, and post-purchase experience can influence satisfaction. Even when the product is acceptable, a poor logistics experience may reduce the likelihood of repeat purchase.

Reviews add qualitative context

Customer reviews capture service problems, packaging issues, unmet expectations, and satisfaction signals that may not appear in transaction data.

Segmentation creates targeting opportunities

Customer clustering and behavioral profiling can help differentiate low-frequency customers, high-value customers, and customers who may require different retention incentives.

Managerial Implication

If shipping cost is a major driver, retention campaigns should not rely only on generic discounts. Shipping incentives, free-shipping thresholds, and loyalty-based delivery benefits may be more aligned with customer pain points.

6. Predictive Modeling

Several classification models were trained to predict customer repeat-purchase behavior. The purpose was not only to identify the algorithm with the highest metric, but to select a model that could provide useful business interpretation and support actionability.

Decision Trees offered interpretability and helped identify the relative importance of variables such as shipping cost and average product price. Random Forest improved predictive robustness by combining multiple trees and reducing the risk of overfitting associated with a single decision tree.

K-Nearest Neighbors and Support Vector Machines were also evaluated. These models helped compare different learning approaches, but Random Forest provided a strong balance between performance, interpretability, and business usability.

For a retention use case, the model should be interpreted as a prioritization tool. Its output can help marketing and customer success teams identify which customers should receive additional incentives or follow-up actions.

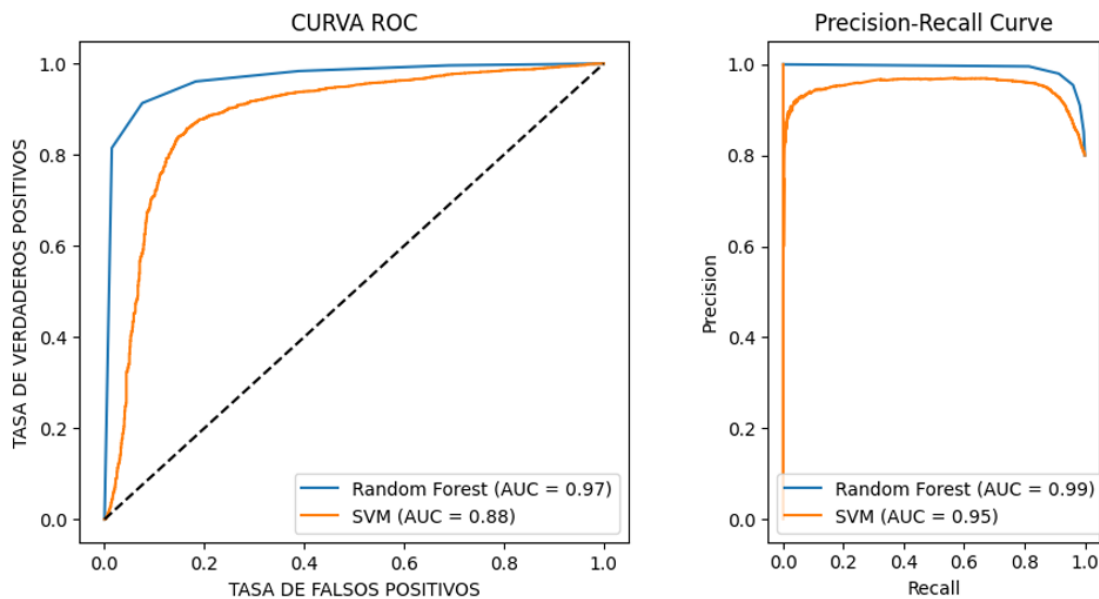


Figure 6. ROC and Precision-Recall curve comparison between Random Forest and SVM.

Dimension	Project Decision
Decision Tree	Useful for interpretability and initial feature importance.
Random Forest	Best balance between predictive performance and business interpretability.
KNN	Useful benchmark based on similarity between customers.
SVM	Strong alternative model for classification comparison.
Recommended model	Random Forest, because it combines solid performance with actionable feature importance.

7. Review Sentiment Analysis

The project also incorporated natural language processing to analyze customer reviews. Reviews were classified into positive and negative sentiment categories based on customer ratings and review text.

This component is important because transactional data can show what customers did, but reviews can explain why they behaved that way. Negative comments may reveal issues related to delivery, packaging, communication, product expectations, or post-purchase support.

From a business perspective, sentiment analysis should be used as an early warning system. Reviews classified as negative can be routed to customer experience teams to identify recurring operational problems and design improvements before they affect future retention.

Dimension	Project Decision
Text source	Customer review comments and ratings.
Business purpose	Identify dissatisfaction signals that may affect retention.
Analytical approach	Deep learning / NLP sentiment classification.
Priority metric	Precision for the Negative class, to reduce false positives when flagging negative reviews.
Business use	Customer experience monitoring and service improvement.

Business Insight

Ratings alone may hide operational problems. Some customers give acceptable scores while still describing issues that can reduce future loyalty. Text analysis helps detect these hidden friction points.

8. Business Recommendations

The following actions translate model outputs into operational retention initiatives. The purpose is not to discount broadly, but to target the right customer with the right incentive at the right moment.

1. Targeted shipping incentives

Offer shipping discounts or free-shipping benefits to customers predicted as high risk, especially when freight cost is high relative to product value.

2. Minimum order threshold for free shipping

Encourage higher basket value by offering free shipping above a specific purchase amount. This can increase average order value while reducing delivery friction.

3. Personalized post-purchase coupons

Send time-limited coupons shortly after the first purchase to encourage a second transaction and reduce the probability of one-time buying behavior.

4. Loyalty program with delivery benefits

Create loyalty tiers that reward repeat purchases with shipping discounts, birthday benefits, exclusive offers, or faster delivery options.

5. Product recommendation campaigns

Use purchase history to recommend complementary products immediately after the first order and during the following weeks.

6. Automated customer communication

Trigger email or WhatsApp campaigns for customers who have not returned after a defined period, using personalized offers and product suggestions.

7. Review-based service recovery

Flag negative reviews and contact customers with unresolved issues. Service recovery can prevent dissatisfaction from becoming permanent churn.

8. A/B testing of pricing and shipping policies

Test alternative combinations of product price and shipping cost to understand which offer structure generates higher conversion and repeat purchase rates.

9. Customer risk scoring dashboard

Create a dashboard for marketing teams showing customers by risk level, predicted retention probability, order value, shipping cost, and recommended action.

10. Continuous model monitoring

Track model performance over time and update the model as customer behavior, logistics costs, and product categories change.

Managerial Implication

The strongest strategy is not one universal discount. The model should support differentiated actions by customer risk level, expected value, and friction points.

9. Business Impact and Future Work

The main business value of this project lies in improving the allocation of retention resources. Instead of applying discounts broadly, Olist could focus incentives on customers with higher probability of churn and higher potential value. This approach can increase marketing efficiency and support customer lifetime value.

The project also connects predictive analytics with managerial decision making. Model outputs can become operational tools for marketing, logistics, customer success, and product teams. For example, a customer risk score could be integrated into CRM workflows or marketing automation platforms.

Future improvements should focus on moving from a portfolio model to a production-ready analytical system. This would require ongoing data updates, model monitoring, A/B testing, customer lifetime value estimation, and integration with business dashboards.

Dimension	Project Decision
Short-term impact	Better targeting of retention campaigns and lower inefficient marketing spending.
Medium-term impact	Improved repeat-purchase rate and higher customer lifetime value.
Long-term impact	Data-driven retention strategy integrated into business operations.
Next analytical step	Add customer lifetime value and campaign response data.
Next business step	Design and evaluate A/B tests for shipping incentives and loyalty benefits.

Future Work

A production version of this project should estimate not only who is at risk, but also which action is most effective for each customer and whether the expected profit justifies the retention incentive.